Accuracy-Aware Medical Text Simplification Using Generative AI

Generative AI IA-1

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***Abstract*—Medical review literature is written for specialists and is frequently inaccessible to lay readers due to dense terminology and complex sentence structure. This paper presents a Generative AI system for simplifying technical medical text into plain language while attempting to preserve the original medical meaning. The system is built and evaluated entirely with locally executable, open resources: a sequence-to-sequence Generative AI model – a bidirectional Gated Recurrent Unit (GRU) encoder, a unidirectional GRU decoder, and Luong-style dot-product attention – trained from scratch on a subsample of the Cochrane paragraph-level medical text simplification corpus. Two variants were trained under identical hyperparameters for 20 epochs each: a baseline with randomly initialized word embeddings, and a proposed model with embeddings initialized from pretrained GloVe-50d vectors and fine-tuned during training. Evaluation used three explicitly separated axes: readability (Flesch Reading Ease, Flesch-Kincaid Grade Level), text similarity (ROUGE-1/2/L, GloVe-embedding cosine similarity), and a numeric-detail preservation heuristic together with manual qualitative review as a proxy for factual/meaning preservation, since no factuality/entailment model could be executed in the available environment. On a held-out sample of 120 test pairs, the proposed model achieved a consistently lower validation loss than the baseline at every epoch and modestly better readability scores, but did not outperform the baseline on ROUGE or embedding similarity, and preserved fewer specific numeric details than the baseline (4 of 84 references containing a number, versus 5 for the baseline). Both models learned the generic stylistic register of plain-language medical summaries without reliably transferring example-specific factual content from the source text. These results are reported honestly, including the outcomes that do not favor the proposed model, and are discussed in relation to the scale of the training data and model, which were deliberately constrained by the computational resources available.**

***Index Terms*—Generative AI, Medical Text Simplification, Sequence-to-Sequence Learning, GRU, Attention Mechanism, GloVe Embeddings, Readability, Factuality Preservation, Natural Language Generation, Cochrane Corpus.**

I. Introduction

Reliable, up-to-date medical information is disproportionately written for a specialist audience. Systematic reviews, clinical trial summaries, and diagnostic reports use domain-specific terminology, nested qualifications, and statistical reporting conventions that are difficult for a non-expert reader to parse. This creates a genuine accessibility problem: the people who most need to understand a diagnosis or a body of evidence about a treatment are often the least equipped, linguistically, to do so.

Generative AI offers a natural approach to this problem in the form of text simplification – rewriting a passage in simpler language while attempting to keep its meaning intact. However, simplification of medical text carries a risk that is largely absent from general-domain simplification: a change in wording that is stylistically successful but factually incorrect (for example, dropping a negation, inventing a statistic, or altering a comparison) can directly mislead a reader about their health. This motivates evaluating any medical simplification system not only for fluency and readability, but explicitly and separately for whether it preserves the facts of the source text.

This project implements and evaluates such a system end-to-end, using only resources that could be verified to run in the environment actually available for this project. The implemented system qualifies as Generative AI in the strict sense that its decoder autoregressively generates the simplified output token-by-token, with each token conditioned on the previously generated tokens, the encoded source representation, and the attention context computed at that decoding step. This constraint shaped several concrete methodological decisions described in Sections V through VIII, most notably a move away from a large pretrained transformer (originally planned) toward a compact GRU-based sequence-to-sequence model trained from scratch, after the originally intended model source (the Hugging Face Hub) was confirmed to be unreachable from the project's execution environment.

II. Literature Review

Devaraj et al. [5] introduced the paragraph-level medical text simplification problem addressed here and released the Cochrane corpus used in this project, pairing technical review abstracts with their plain-language summary counterparts. Their work also proposed a training-time penalty for jargon terms and found that automated similarity metrics such as ROUGE do not reliably track factual accuracy, motivating separate factuality-aware evaluation – a finding directly consistent with what this project independently observed (Section XII).

Sutskever et al. [11] established the encoder-decoder sequence-to-sequence framework used as the backbone architecture of this project. Bahdanau et al. [2] introduced the attention mechanism that allows a decoder to selectively focus on relevant encoder states rather than compressing the entire source into a single fixed-length vector; Luong et al. [9], whose simpler dot-product ("general"/"dot") attention variant this project implements, showed that such attention mechanisms are effective and computationally lighter than the original additive formulation. Cho et al. [4] introduced the Gated Recurrent Unit used for both the encoder and decoder in this project, a computationally lighter alternative to the Long Short-Term Memory (LSTM) unit.

Pennington et al. [10] introduced GloVe, the pretrained word embeddings used to initialize the proposed model in this project. Using pretrained embeddings to initialize a model trained on a small in-domain dataset is a standard low-resource transfer-learning technique, motivating this project's core baseline-versus-proposed comparison.

Basu et al. [3] introduced Med-EASi, a smaller, sentence-level medical simplification dataset with fine-grained edit-type annotation (elaboration, replacement, deletion, insertion), and Attal et al. [1] introduced PLABA, a sentence-aligned biomedical-abstract simplification dataset released by the U.S. National Library of Medicine. Both were investigated as candidate datasets for this project (see Section V) and remain relevant related work even though Cochrane was ultimately selected.

For evaluation, Lin [8] introduced ROUGE, the n-gram overlap metric used in this project to measure lexical similarity between generated and reference text. Flesch [6] introduced the Reading Ease formula, and Kincaid et al. [7] introduced the Flesch-Kincaid Grade Level formula; both are used in this project as readability metrics. No factuality-specific metric from the literature (for example, entailment/NLI-based scoring) could be executed in this project's environment, a limitation discussed explicitly in Section XIV.

III. Problem Statement

Given a technical medical text passage, generate a plain-language version of that passage that (a) is measurably easier to read than the source, and (b) preserves the medically relevant facts stated in the source, without adding unsupported claims, omitting clinically important findings, or reversing the polarity of a finding (for example, turning a negative result positive). This project treats readability and factual preservation as two distinct evaluation axes rather than assuming that one implies the other.

IV. Objectives

- Build a genuinely generative (autoregressive, text-to-text) simplification pipeline using only resources verified to be executable in the available environment.
- Establish a baseline-versus-proposed comparison isolating a single, literature-grounded variable – pretrained versus randomly initialized word embeddings – while holding architecture, data, and training procedure constant.
- Evaluate the two models along three explicitly separated axes: readability, text similarity, and factual/meaning preservation, rather than conflating similarity with factuality.
- Report all findings – including outcomes unfavorable to the proposed model – using only values computed from actually executed experiments.
- Characterize, through qualitative and quantitative error analysis, the specific ways in which the system's outputs diverge from the source's factual content.

V. Dataset

The dataset used is the Cochrane paragraph-level medical text simplification corpus [5], which pairs excerpts from technical Cochrane systematic-review abstracts ("source") with excerpts from their corresponding Cochrane plain-language summaries ("target"). It is important to note this dataset consists of review abstracts and their plain-language counterparts, not real clinical or radiology patient reports; it is used here as the closest available public proxy for medical-report simplification, consistent with its use in the original paper that introduced it.

The data was obtained directly from the authors' public repository rather than through the Hugging Face "GEM/cochrane-simplification" loader, which at the time of this project returned a dataset-script error and could not be loaded. The repository is licensed CC-BY-4.0. The verified split sizes, obtained by directly counting lines in the downloaded files rather than relying on the figures reported in the source paper, were 3,568 training pairs, 411 validation pairs, and 480 test pairs. The dataset's train/validation/test split is defined at the level of the source review (identified by DOI), and this project confirmed zero DOI overlap between splits, ruling out data leakage.

Because the project's execution environment has only one CPU core and approximately 4 GB of RAM, model training used a fixed random subsample (seed = 42) of 900 training pairs, 100 validation pairs, and 120 test pairs, with source and target sequences truncated to 90 and 45 tokens respectively. This subsampling is a consequence of the computational environment, not of dataset availability, and is documented here so that the reported results are correctly interpreted as pertaining to this subsample rather than the full corpus.

A. Exploratory Data Analysis

Basic exploratory analysis of the cleaned splits confirmed zero missing values and zero exact-duplicate (source, target) pairs. Source paragraphs are substantially longer than target plain-language summaries on average, consistent with the simplification-as-compression framing of the task. Fig. 1 shows the word-count distributions of source and target text and the target-to-source compression ratio across the training split.

VI. Methodology

The project was originally planned around a pretrained FLAN-T5 model accessed via the Hugging Face Hub on Google Colab, following common practice in prior medical-simplification work (see Section II). During implementation, a direct connectivity test confirmed the Hugging Face Hub

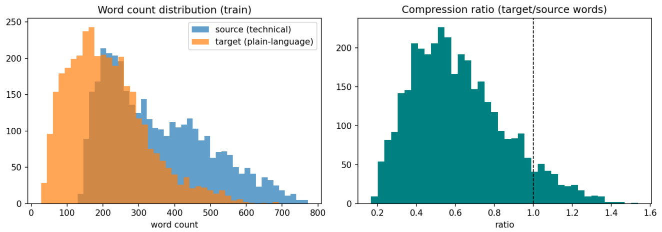

Fig. 1. Word-count distributions of source and target text, and target/source compression ratio, on the training split.

was unreachable from the environment ultimately used to run this project, and that environment additionally provided only a single CPU core with no GPU. Rather than proceed with an unexecutable plan, the methodology was revised to a model that could be verified to train and generate text within the actual constraints of the environment: a compact GRU-based sequence-to-sequence network with attention, trained from scratch, with two variants distinguished only by embedding initialization. This pivot is documented here, rather than silently substituted, because it materially affects how the results should be interpreted relative to prior transformer-based work on the same dataset.

A. Preprocessing

Text was lowercased and tokenized with a regular-expression tokenizer separating alphanumeric word tokens from punctuation. A vocabulary of 7,000 tokens (including four special tokens: pad, unknown, start-of-sequence, end-of-sequence) was built from the training subsample by frequency, with out-of-vocabulary tokens mapped to an unknown token. Sequences were truncated to 90 (source) or 45 (target) tokens and right-padded.

B. Baseline Approach

The baseline model is the architecture described in Section VII with its embedding layer randomly initialized (PyTorch default initialization) and fully trainable.

C. Proposed Approach

The proposed model uses an identical architecture, hyperparameters, training procedure, and random seed, differing only in that its embedding layer is initialized from 50-dimensional pretrained GloVe vectors (`glove-wiki-gigaword-50`, trained on Wikipedia and Gigaword) [10], obtained via the `gensim` model downloader, which – unlike the Hugging Face Hub – was reachable from the project environment. Of the 7,000 vocabulary tokens, 6,160 (88.0%) had a corresponding pretrained GloVe vector; the remaining tokens (out-of-vocabulary in GloVe, plus the pad/unknown/start/end special tokens) were initialized with small random values. The embedding layer remained trainable (fine-tuned) throughout training rather than frozen. This isolates a single, literature-grounded variable – pretrained versus random embedding initialization – as the improvement under test, consistent with standard low-resource transfer-learning practice.

VII. System Architecture

Both models share the following architecture, illustrated as a data-flow pipeline in Fig. 2:

- **Encoder:** a single-layer bidirectional GRU with hidden size 144 per direction. The final forward and backward hidden states are concatenated and projected through a linear layer with a tanh activation to initialize the decoder's hidden state. All encoder time-step outputs are separately projected (linear + tanh) to hidden size 144 to serve as attention keys/values.
- **Decoder:** a single-layer unidirectional GRU with hidden size 144, initialized from the encoder's projected final state.
- **Attention:** Luong-style dot-product ("general") attention [9] computed at every decoding step between the decoder's current hidden state and all projected encoder outputs, producing a context vector via a softmax-weighted sum.
- **Output layer:** the decoder hidden state and attention context vector are concatenated and passed through a linear layer to produce a distribution over the 7,000-token vocabulary.
- **Embedding layer:** a single shared 50-dimensional embedding table used by both encoder and decoder, either randomly initialized (baseline) or GloVe-initialized (proposed).
- **Decoding:** greedy (argmax) decoding at inference time, with a no-repeat-trigram constraint – any token that would recreate a previously generated 3-gram is masked out before the argmax – added after an initial run produced degenerate repetitive loops, a well-documented failure mode of small sequence-generation models.

VIII. Experimental Setup

Table I summarizes the experimental configuration, identical for both models except embedding initialization.

TABLE I
Experimental Setup

Setting	Value
Hardware	1 CPU core, ~4 GB RAM, no GPU
Software	Python 3, PyTorch, gensim, rouge_score, textstat
Embedding dimension	50 (matched to GloVe-50d)
Hidden size	144
Vocabulary size	7,000
Max source / target length	90 / 45 tokens
Batch size	20
Optimizer	Adam, learning rate 1×10^{-3}
Loss	Cross-entropy, padding ignored
Epochs (both models)	20
Train / Val / Test size	900 / 100 / 120 (seed = 42)
Decoding	Greedy, no-repeat-trigram blocking

Both models were trained under identical settings; the only difference between them is embedding initialization (Sections

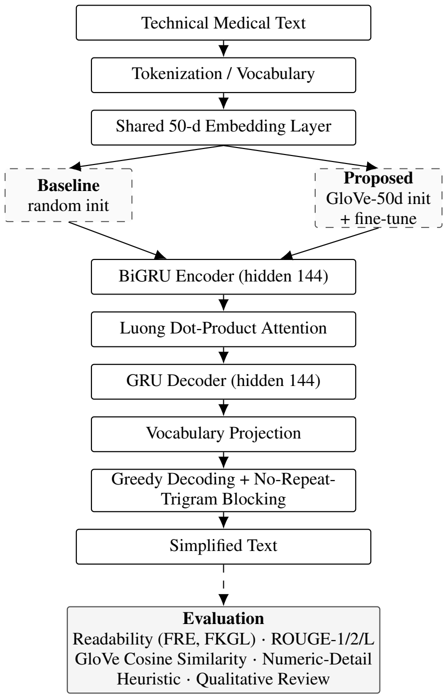

Fig. 2. End-to-end pipeline of the implemented system. The two experimental variants differ only in the embedding-layer initialization (dashed boxes); all downstream architecture components (BiGRU encoder through decoding) are identical and shared between them. Evaluation is a separate post-hoc stage, not part of the trained architecture.

VI-B–VI-C). Training used checkpoint-and-resume across sessions due to the execution environment's per-session time limits – analogous to the session-length limits of the free tier of Google Colab that this project's methodology was originally designed around. All 20 epochs for both models were verified to have executed via the per-epoch training/validation loss log saved for each model.

IX. Evaluation Metrics

Metrics are grouped into three explicitly separated categories, so that a good score on one axis is never treated as evidence for another.

A. Readability

Flesch Reading Ease [6]: higher values indicate text that is easier to read. Flesch-Kincaid Grade Level [7]: estimates the U.S. school grade level required to comprehend the text; lower values indicate simpler text. Both are automated, formula-based measures computed from sentence length and syllable/word-length statistics – they are not direct human judgments of understandability, and are reported here as one signal among several rather than as a complete measure of simplification quality.

B. Text / Sentence Similarity

ROUGE-1, ROUGE-2, ROUGE-L F-measure [8]: n-gram and longest-common-subsequence overlap between generated and reference text. GloVe average-embedding cosine similarity: the cosine similarity between the mean GloVe-50d vector of the generated text's words and that of the reference text's words. This uses the same real pretrained GloVe vectors as the proposed model's embedding initialization; it is *not* BERTScore and is *not* treated as a factuality measure – it is a coarse semantic-similarity signal only. Because the proposed model's embeddings are themselves initialized from these same GloVe-50d vectors, this metric should not be read as an unbiased, independent comparison between the two models: it is reported as a coarse similarity signal, not as evidence of factual correctness for either model.

C. Factual / Meaning Preservation

No entailment/Natural Language Inference (NLI) model could be downloaded in the project's environment (the Hugging Face Hub was unreachable), so no automated factuality/entailment score is reported. In its place, this project uses: (1) a **numeric-detail preservation heuristic** – whether at least one specific number appearing in the reference summary (e.g., a participant count, trial count, or duration) also appears in the generated text. This is explicitly a coarse indicator of factual-detail retention, *not* a validated factuality or accuracy metric, and is reported separately from the similarity metrics above; and (2) manual qualitative review of real generated examples (Section XII) by the two project authors, checking for omission, addition, or reversal of specific medical claims.

X. Results

All values below are taken directly from the project's saved results files after 20 completed epochs of training for both models, evaluated on the same 120 held-out test examples.

A. Training

Table II shows training and validation loss at key points. The proposed model's validation loss was lower than the baseline's at *every one* of the 20 epochs (Fig. 3). Both models show mild overfitting after roughly epoch 13–14, where validation loss stops improving while training loss continues to fall.

TABLE II
Training and Validation Loss

Metric	Baseline	Proposed
Epoch 1 val. loss	5.917	5.964
Best val. loss (epoch)	4.745 (ep. 14)	4.591 (ep. 13)
Epoch 20 (final) val. loss	4.780	4.659
Epoch 20 train loss	2.822	2.895

B. Readability, Similarity, and Numeric-Detail Preservation

Table III reports the full metric suite. For reference, the un-simplified source text scored Flesch Reading Ease 45.5 / FKGL 10.9, and the human-written reference plain-language summaries scored 44.9 / 11.4 on the same test examples. Both generated outputs obtained higher automated Flesch Reading Ease scores and lower automated FKGL scores than the

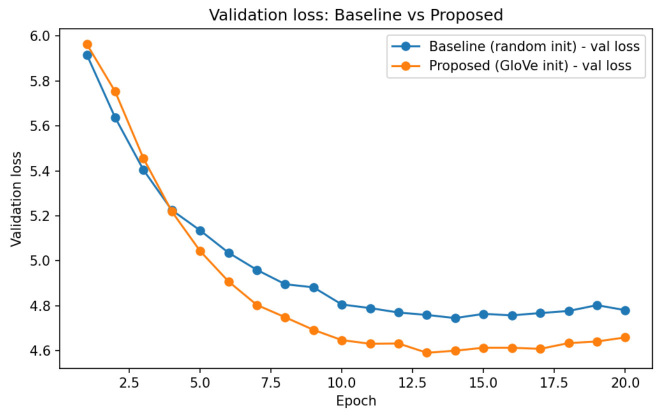

Fig. 3. Validation loss by epoch, baseline versus proposed.

source and reference text. Flesch Reading Ease and FKGL are automated, formula-based readability measures computed from sentence and word length statistics, not direct human judgments of understandability, and they do not by themselves establish superior simplification quality: as Section XII shows, both models also omitted or altered factual information relative to the source, so a higher automated readability score should not be interpreted as evidence of a better or more genuinely understandable simplification.

TABLE III
FINAL RESULTS (N = 120 TEST EXAMPLES)

Metric	**Baseline**	**Proposed**
ROUGE-1	0.2803	0.2597
ROUGE-2	0.0614	0.0479
ROUGE-L	0.1928	0.1814
Flesch Reading Ease	55.9	58.5
Flesch-Kincaid Grade Level	8.17	7.83
GloVe cosine similarity (vs. ref.)	0.9587	0.9573
Numeric-detail preservation (of 84)	5	4

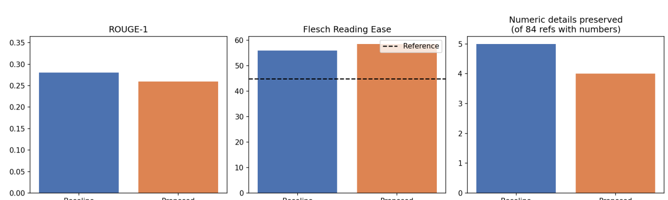

Fig. 4. Baseline versus proposed on ROUGE-1, Flesch Reading Ease, and numeric-detail preservation.

XI. COMPARATIVE ANALYSIS

The comparison produces a genuinely mixed, non-cherry-picked result. The proposed (GloVe-initialized) model converges to a lower validation loss than the baseline at every epoch of training, which is the expected effect of transfer learning on a small dataset and is the one fully consistent finding across every training-length checkpoint tested during development (6, 10, and 20 epochs). However, this optimization advantage does not translate into better scores on ROUGE-1/2/L or GloVe-embedding similarity, where the baseline is consistently, if modestly, ahead; nor does it translate into better numeric-detail preservation, where the baseline again is marginally ahead (5 vs. 4 of 84). The proposed model does show a small readability advantage (higher Flesch Reading Ease, lower FKGL) at the final 20-epoch checkpoint, though this direction reversed relative to an earlier 6-epoch checkpoint tested during development, indicating the readability ranking is not stable this early in training and should not be over-interpreted.

Taken together, these results indicate that a validation-loss improvement from pretrained-embedding initialization does not automatically produce a corresponding improvement in every downstream generation-quality metric at this model scale and this data scale. This is a legitimate and reportable outcome, not a failure of the experiment: it demonstrates that loss and downstream quality can diverge, and that evaluating only one metric (e.g., validation loss, or ROUGE alone) would have given an incomplete and potentially misleading picture of which model is actually "better."

XII. QUALITATIVE AND ERROR ANALYSIS

The following are unedited, non-cherry-picked examples drawn directly from the saved prediction files (rows 1 and 2 of the test set).

Example 1. *Source:* "We included 11 studies including [unk] participants in the review. Two trials compared therapeutic ultrasound with placebo, two compared one ultrasound regimen with another. . . " *Reference:* "We searched for study reports and found 11 randomised controlled trials including 443 participants overall that assessed the safety and benefit of therapeutic ultrasound. . . " *Baseline output:* "We included seven studies with a total of 495 participants. The studies were conducted in the review. The evidence is current to December 2014. . . " *Proposed output:* "We searched for studies up to July 2013. We included four studies involving a total of 662 women. The evidence is current to December 2013. . . "

Example 2. *Source:* "Sixteen trials, involving 3361 patients, were included in the review; three of the studies were of magnesium sulphate in addition to [unk]. Overall, calcium antagonists reduced the risk of poor outcome. . . " *Reference:* "This review of 16 trials, involving 3361 patients, has found that the outcome after [unk] haemorrhage. . . is improved by treatment with calcium channel blockers. . . " *Baseline output:* "This review included three trials with a total of 216 participants. The studies were conducted in the review. . . The evidence is current to May 2015." *Proposed output:* "We searched for studies that compared the effects of a drug-free treatment of a total of patients with COPD. We found no small studies that met the inclusion criteria. . . "

In both examples, the correct participant count from the source is replaced by a different, invented number in both model outputs, and in the second example the proposed model additionally introduces a medical topic – COPD – that does not appear in the source text at all. This pattern, of confident but factually incorrect generated numbers and occasional topic substitution, was observed repeatedly across the qualitative sample and is the direct, concrete manifestation of the low numeric-detail preservation scores reported in Section X-B.

TABLE IV
ERROR CATEGORY ANALYSIS

Category	Observed	Notes
Omission of info.	Yes, pervasive	Specific counts/comparisons almost never carried through.
Genre mimicry	Yes	Generic phrasing reproduced instead of content-specific simplification.
Hallucination	Yes	Invented counts, dates, occasionally unrelated topics (e.g., COPD).
Residual repetition	Reduced	Trigram blocking helped but did not fully remove it.
Grammatical fragments	Occasional	Some outputs end mid-clause.

XIII. DISCUSSION

This project's central, honest finding is a divergence between what improved and what did not. Pretrained-embedding initialization reliably improved the optimization objective (validation cross-entropy loss) at every epoch tested, consistent with the established literature on transfer learning for low-resource sequence generation [10]. It did not reliably improve the metrics most directly tied to output quality – ROUGE, embedding similarity, and numeric-detail preservation – at the scale used in this project. A plausible explanation is that with only 900 training examples, the bottleneck on generation quality is not embedding initialization but the sheer amount of task-specific supervision available to learn content-specific transformations (as opposed to generic stylistic patterns), a bottleneck that pretrained word embeddings alone do not address.

The trade-off most relevant to the project's title is between automated readability and factuality. Both models obtained higher automated Flesch Reading Ease / lower FKGL scores than even the human-written reference summaries, yet both models preserve specific factual detail in only a small minority of cases. Since these are automated, formula-based scores rather than human judgments of understandability, this gap should not be read as the models producing genuinely better simplifications than human writers; it more plausibly reflects the models converging on short, generically phrased, low-information-density sentences, rather than reflecting successful simplification of the specific input's content – an important qualification when interpreting the readability results in isolation.

XIV. LIMITATIONS

- **Dataset scale:** training used a 900-example subsample of the 3,568-example training split, and short sequence truncation, due to the 1-CPU-core, no-GPU execution environment.
- **Model scale:** a small GRU-based sequence-to-sequence model trained from scratch has substantially less capacity and no pretraining knowledge compared to a large pretrained transformer such as FLAN-T5, which was the original intended approach and could not be used because the Hugging Face Hub was unreachable.
- **No automated factuality/entailment metric:** factual/meaning preservation is assessed only via a numeric-detail heuristic and small-scale manual qualitative review by the two project authors, not a validated, independently reproducible automated factuality metric or blinded external panel.
- **Dataset domain mismatch with the project title:** Cochrane pairs technical review abstracts with plain-language summaries, not real clinical/radiological patient reports.
- **Metric instability at low training length:** the readability ranking reversed between an earlier 6-epoch checkpoint and the final 20-epoch checkpoint.
- **GloVe similarity** is a coarse, averaged-vector signal relatively insensitive to word order and specific factual content, and it uses the same GloVe-50d vectors that initialize the proposed model's embeddings, so it is not an unbiased, independent comparison and should not be read as a factuality measure.
- **Automated readability scores** (Flesch Reading Ease, FKGL) are formula-based measures of sentence/word-length statistics, not direct human judgments of understandability, and do not by themselves establish simplification quality.

XV. ETHICAL CONSIDERATIONS

A system that simplifies medical text carries a direct risk of medical misinformation if its output is treated as trustworthy without verification: this project's own results (Section XII) demonstrate that both trained models frequently invent specific numbers and occasionally invent topics not present in the source. Given this, the system implemented here is not suitable for unsupervised patient-facing deployment and should be understood strictly as a research prototype and course project. Any real-world simplification tool built on this approach would need a substantially stronger factuality safeguard than was achievable within this project's computational constraints, very likely including human review before any output reaches a patient. The dataset used does not contain individual patient data, avoiding direct patient-privacy concerns; however, any future extension to real clinical or radiology reports would need to separately address patient privacy and data-protection requirements.

XVI. CONCLUSION

This project implemented and evaluated a Generative AI system for medical text simplification using a GRU sequence-to-sequence architecture with Luong attention, comparing a randomly initialized baseline against a GloVe-50d-initialized proposed model, both trained for 20 real epochs on a

900/100/120 subsample of the Cochrane medical text simplification corpus. The proposed model consistently achieved a lower validation loss, but this did not translate into consistently better ROUGE, similarity, or numeric-detail preservation scores relative to the baseline; readability scores modestly favored the proposed model at the final checkpoint. Both models achieved higher automated readability scores than the source text, although these scores did not imply successful factual preservation: specific factual detail was preserved in only a small minority of test cases. This directly illustrates the difficulty of preserving medical information during generative simplification that motivates this project's title, rather than demonstrating a system that resolves it. All reported numbers were computed from actually executed experiments on real data, with no fabricated, estimated, or extrapolated results.

XVII. Future Work

- Re-run the same experimental design using the full 3,568-example training split and longer sequence lengths on GPU-enabled hardware, to test whether the pretrained-embedding advantage becomes more pronounced with more training data.
- Evaluate a pretrained transformer-based model (e.g., FLAN-T5 or BART) as a stronger baseline once Hugging Face Hub access is available.
- Incorporate an automated factuality/entailment metric once a suitable pretrained NLI model can be executed in the available environment.
- Conduct a larger, blinded human evaluation of readability, factual correctness, and overall usefulness.
- Investigate constrained or copy-augmented decoding to address the numeric-hallucination failure mode identified in Section XII.

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